NAME: _____ Geometry (FL B.E.S.T.)

10.1 Surface Area

Surface area (SA) is the total area of every face of a 3D solid — the amount of material needed to wrap it completely. Prisms and cylinders use $SA = 2B + Ph$ (or the equivalent $2(lw + lh + wh)$ for a box), pyramids and cones use a base plus *lateral* area built from the **slant height** ℓ (not the vertical height), and $SA = 4\pi r^2$ for a sphere. Many real problems hide a Pythagorean-theorem step: the slant height of a cone or pyramid is rarely given outright — you often build it from the height and a base measurement first. This worksheet also covers composite solids (two shapes fused, with the shared seam excluded) and reverse problems where SA is known and a dimension is missing.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Find the SA of a rectangular prism with length 6, width 4, height 5.

2. Find the SA of a cube with edge 9.

3. Find the total SA of a cylinder with radius 5, height 11.

4. Square pyramid, base edge 8, slant height 9. Find total SA.

5. Cone with radius 9, slant height 15. Find total SA.

6. Find the SA of a sphere with radius 7.

7. Find the SA of a sphere with diameter 20.

8. Cone with radius 8, height 6. Find the slant height, then the total SA.

9. Square pyramid, base edge 14, height 24. Find the slant height, then the total SA.

10. A silo is a cylinder (radius 4, height 9) topped with a hemisphere of the same radius. Find the total outer SA (no bottom, no seam).

11. A rectangular prism has length 8, width 5, and SA 262. Find its height.

12. An open-top cylindrical tank has radius 6, height 12. Find the SA of material needed (no lid).

13. Explain why a square pyramid's total SA formula adds only *one* base area, while a prism's formula effectively adds two.

14. A silo (cylinder radius 5, height 14, topped by a hemisphere of the same radius) needs to be painted on the outside, excluding the bottom (in the ground). Find the area to be painted.

15. A cone and a cylinder share the same base radius r. The cylinder has height h and the cone has slant height ℓ. If their total surface areas are equal, show that $\ell = r + 2h$.

✓ Answer Key

1. 148
2. 486
3. 160π
4. 208
5. 216π
6. 196π
7. 400π
8. $\ell = 10$; $SA = 144\pi$
9. $\ell = 25$; $SA = 896$
10. 104π
11. $h = 7$
12. 180π
13. A pyramid has a single base; a prism has two congruent parallel bases
14. 190π
15. $2\pi r(r + h) = \pi r(r + \ell) \Rightarrow 2(r + h) = r + \ell \Rightarrow \ell = r + 2h$